

(302) 772-9488
denardiluc@gmail.com
lucde4390@colorado.edu

Luc De Nardi

Portfolio: look-luc.github.io/luc_portfolio
github.com/look-luc
linkedin.com/in/luc-de-nardi

PROFESSIONAL SUMMARY

Solution focused computational linguist with experience in NLP pipelines, transformer/classification models, and data processing workflows. Applying linguistic knowledge into the realm of machine learning to expand to the under documented languages sphere while exploring how to use current AI applications.

EDUCATION

University of Colorado Boulder

August 2025 – Present

Master of Science in Computational Linguistics (CLASIC), Advisor: Dr. Susan Brown

University Delaware

May 2025

Bachelor of Arts in Linguistics with Minors in Computer science and Cognitive Science

TECHNICAL SKILLS

Programming/Platforms	Java, Python (Pytorch, NLTK, Pandas, HuggingFace), Elan, Javascript/TypeScript, C/C++, Terminal (VIM, Bash), Git (Github & Gitlab), LaTeX, PCIBex, Praat, and SQL (PostgreSQL and Oracle SQLDeveloper), and Django
Tools	Software development processes; Music composition, sound editing, 3D modelling (Blender), and CAD design (Fusion 360)
Research & NLP	Low-resource language modeling, tokenization analysis, speech processing, linguistic annotation, experimental design, data parsing
Languages	English (Native), French (intermediate), Spanish (pre-intermediate), and Arabic (novice)

RESEARCH ASSISTANT EXPERIENCE

Speculative Decoding, University of Colorado Boulder [PI Dr. Alexis Palmer]

November 2025 – May 2026

- Advanced research in low-resource language generation by developing a speculative decoding draft model that used monolingual and bilingual corpora, enabling a more efficient translation and story-generation workflows while supporting experimentation in under-resourced language settings

French Clear Speech, University of Colorado Boulder [PI Dr. Rebecca Scarborough]

October 2025 – Present

- Improved analysis of speech variation by designing and training a random-forest classification model that identified acoustic features distinguishing speech styles within targeted phonetic categories
- Designed a study, utilizes a pretrained model to act as a listener, that will gather more data for identifying acoustic features distinguishing speech styles within targeted phonetic categories

Bantu Proverb Literal Translation, University of Colorado Boulder/University of Minnesota-Twin Cities

July 2026 – Present

- Evaluating quality of literal proverb translations across multiple Bantu languages by implementing linguistic glossing and native speaker annotations comparisons
- Managed end-to-end data acquisition workflows by coordinating with native speakers to collect authentic translations, ensuring data is high-quality for later use

Co-Speech Gesture, Phonlab University of Delaware/[Harvard University](#)

February 2022 – February 2025

- Increased the reliability and usability of multimodal linguistic datasets by preprocessing hand and head gesture video recordings and applying standardized coding protocols, supporting downstream analysis of co-speech gesture behavior
- Contributed to ongoing cross-linguistic gesture research by expanding annotated datasets and produced accurate transcriptions of Cameroonian English, enabling broader analysis of speech-gesture interactions across diverse language varieties

Semantic vs Phonological Patterns & Perception in Object Symmetry, [UD PAL Lab](#) University of Delaware

[PI Dr. Alon Hafri]

February 2023 – May 2023

- Investigated relationships between verb semantics and phonological patterns across languages, developing Python-based data analysis methods to evaluate whether specific phonemic features correspond to abstract conceptual distinctions
- [Supported perception research](#) by helping design and implemented a [pilot experiment](#) combining visual and auditory stimuli, established an initial framework for evaluating how object symmetry influences human perception

Programming, Experimental Syntax Lab University of Delaware [PI Dr. Rebecca Tollan]

February 2023 – May 2023

- Assistant programmer, priming test code for processing of ellipses in English

PUBLICATIONS & PRESENTATIONS

Manodyna K H and **Luc De Nardi**. 2026. When Multilingual Evaluation Assumptions Fail: Tokenization Effects Across Scripts. In Proceedings of the Second Workshop on Language Models for Low-Resource Languages (LoResLM 2026), pages 41–49, Rabat, Morocco. Association for Computational Linguistics.

Manodyna K H and **Luc De Nardi**. 2026. Rethinking Polarity Detection: When BPE Fails Across Scripts. In Proceedings of the 2nd Workshop on NLP for Languages Using Arabic Script, pages 6–14, Rabat, Morocco. Association for Computational Linguistics.

Paudel, N., Ginn, M., **De Nardi, L.**, & Palmer, A. (2026). Speculative Decoding Across Languages. arXiv. <https://arxiv.org/abs/2605.30580> (preprint)

PROJECTS

Speculative Decoding Capstone, University of Colorado Boulder

August 2026 – Present

- Experimenting with iterations on speculative decoding by benchmark-testing alternative architectures and decoding strategies, driving performance improvements over prior baseline models.

Spoken English Classifier, University of Colorado Boulder

January 2026 – May 2026

- Engineered an end-to-end NLP pipeline using Conv-BERT to classify discourse overlap from the Santa Barbara Corpus of Spoken American English dataset

Mental Health Resource Search & Taxonomy

May 2026

- Full stack developer & designer of a searchable relational database utilizing Postgresql to filter resources for a Peer Support initiative with Django as a full-stack pipeline

WORK EXPERIENCE

2025 SAFAL Conference [PI Dr. Bhuvana Narasimhan]

Technical student organizer

- Technical student organizer, managing technical logistics during conference, facilitating Zoom, asynchronous presentations, and updated website

ACTIVITIES AND INTERESTS

Sean's House Peer24 Program

- Peer-Specialist volunteer, certified for mental health support of fellow students

Music Ensemble at UDel & CU Boulder

- CU Boulder Campus Orchestra & Boulder Samba School; UD Collegiate Band, UD 8-Bit orchestra, UD Marching band, UD Steel Drum Orchestra, UD F Horn Ensemble, Música Latina

REFERENCES

Upon request.